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BIOLOGY

1. Regarding bacteria:

A. Cocci are rod-shaped

B. Bacilli are spiral shaped

C. All contain RNA

D. None of these

2. Chlamydomonas:

A. Is a multicellular alga

B. Has a single flagellum

C. Has no system of water regulation

D. Has a light sensitive structure

3. Cryptogams:

A. Are non-flowering plants

B. Form seeds+fruits

C. Include conifers+cycads

D. Include alga

4. Molluscs have all of the following EXCEPT:

A. A well developed nervous system

B. Kidneys

C. Nostrils for respiration

D. Circulatory system with a heart

5. Crustacea includes all of the following EXCEPT:

A. Crabs

B. Lobsters

C. Oysters

D. Crayfish

6. The dark reaction of photosynthesis:

A. Takes place in stroma of chloroplast

B. Involves the fixing of water

C. Causes formation of lactic acid

D. Takes place in the cytoplasm

7. The pituitary gland secretes all of the following EXCEPT:

A. Growth hormone

B. Calcitonin

C. Prolactin

D. Oxytocin

8. Thyroid stimulating hormone levels:

A. Would be raised in person with hyperthyroidism

B. Would be raised in patient with hypothyroidism

C. Would be low in patient with hypothyroidism

D. Would be unaffected in hypothyroidism

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9. Gastrulation in a frog's embryo involves all of the following EXCEPT:

A. Epiboly

B. Involution

C. Rotation

D. Invagination

10. Mesoderrm gives rise to:

A. Lens of the eye

B. Teeth enamel

C. Liver

D. Heart

11. Gaseous exchange in planaria takes place by means of:

A. Osmosis

B. Diffusion

C. Breathing

D. Active transport

12. Retina contains light sensitive receptor cells called:

A. Rods

B. Cones

C. Both rods and cones

D. Optic nerves

13. Excretory system of grasshopper consists of:

A. Nephridia

B. Flame cells

C. Malpighian tubules

D. Kidneys

14. Which of the following Cnidarian lives in colonies?

A. Tapeworm

B. Obelia

C. Jelly Fish

D. Sea anemone

15. The causative agent of malaria is:

A. Plasmodium

B. Trypanosoma

C. Volvox

D. Entamoeba Histolytica

16. Earthworms feed on:

A. Decaying vegetation

B. Organic matter

C. Soil

D. All of the above

17. Planaria feeds through the:

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A. Mouth

B. Pharynx

C. Esophagus

D. Oral groove

18. Sugar is stored in liver as:

A. Starch

B. Glycogen

C. Maltose

D. Cellulose

19. Function of "tube feet" in phylum Echinodermata is:

A. Locomotion

B. Anchoring to hard surface

C. Grabbing the prey

D. All of the above

20. What is the main role of vitamin K in the diet of humans?

A. To participate in the formation of visual purple

B. To promote the production of blood clotting factors

C. To prevent skin disorders

D. Deposition of calcium in the bones

E. All of the above

21. Removal of thyroid gland in an adult would cause an increase in:

A. BMR

B. Conversion of glycogen to glucose

C. Excretion of Na+ from kidneys

D. Secretion of TSH

22. The following is a part of the female reproductive tract:

A. Vas deferens

B. Cervix

C. Spermatozoa

D. Ovary

23. The specificity of enzyme is due to their

A. High molecular weight

B. Hydrogen bonding

C. pH sensitivity

D. Surface configuration

24. The complementary mRNA for the DNA triplet GAT would read:

A. CTA

B. CUA

C. CTG

D. CTC

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25. Which of the following is an example of discontinuous variation?

A. Range if height of pea plants: 30-50cm

B. Offspring including 2 males and 3 females

C. Pea plants grown in darkness are yellow

D. Adult human weight ranges from 50-95 kg

26. The approx. width of a cell membrane is:

A. 0.001nm

B. 0.10nm

C. 1.0nm

D. 10nm

27. Which of the following is not a part of seed?

A. Plumule

B. Pericarp

C. Testa

D. Cotyledon

28. Amoeba causes

A. Dysentery

B. Malaria

C. Sleeping sickness

D. Gangrene

29. An action potential in a muscle fiber causes the release of calcium ions from:

A. Actin

B. Myosin

C. Sarcolemma

D. Sarcoplasmic reticulum

30. All of the following are examples of fungi EXCEPT:

A. Yeast

B. Mushroom

C. Moulds

D. Vorticella

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CHEMISTRY

31. Rutherford discovered that the nucleus of an atom has _____ charge.

A. No

B. Negative

C. Positive

D. Very less

32. How many grams of NaCl are required to completely neutralize 100 ml of 1M of Sulphuric acid.

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A. 80 grams

B. 40 grams

C. 24 grams

D. 8 grams

33. Cyclohexane has the formula:

A. C6H14

B. C6H12

C. C6H16

D. C6H6

34. In

H2C=CH2

the molecule contains:

A. 4 sigma - 2 pi Bonds

B. 2 sigma - 4 pi bonds

C. 5 sigma - 1 pi bonds

D. 6 sigma bonds

35. What is the general formula of Ester?

A. R—COO—R

B. R—CHO

C. R—OH

D. RX

36. The transition elements in the periodic table are classified as:

A. S-Block elements

B. P-Block elements

C. d-Block elements

D. d and f-Block elements

37. Al2O3.H2O

Is called:

A. Bauxite

B. Gyolele

C. Alum stone

D. Alonite

38. What is the boiling point of Nitric Acid?

A. 86°C

B. 40°C

C. 76°C

D. 66°C

39. One mole of Ethanol and one mole of Ethane have an equal:

A. Mass

B. Number of atoms

C. Number of electrons

D. Number of molecules

40. Alkenes undergo:

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A. Addition reactions

B. Substitution reactions

C. Both addition and substitution

D. None of the above

41. During the electrolysis of H2O, H2 is liberated at:

A. Anode

B. Cathode

C. Diode

D. None of the above

42. An example of covalent hydride is:

A. NaH

B. CaH2

C. SrH2

D. AsH3

43. In this Equilibrium Ksp is Expressed as:

BaSO4 harr Ba(+2) + SO(4)(2-)

A. Ksp= BaSoO4

B. Ksp=(Ba(2+)SO(4)(2-))/(BaSO4)

C. Ksp=Ba(2+)SO(4)(2-)

D. None of the above

44. The pH of a solution contains 10^{-4} M [H+] is:

A. 4

B. 3

C. -3

D. 0

45. "The energy can neither be created nor be destroyed but can be transferred and transformed." This is known as:

A. Law of conservation of energy

B. Law of conservation of mass

C. Law of conservation of momentum

D. Hess's law

46. At what temperature standard enthalpy changes are measured?

A. 298K

B. 273K

C. 293°C

D. 298°F

47. Rb is an element similar to potassium which can showbthe decomposition of Rubidium nitride as:

A. 4RbNO3 → 2Rb2O+4NO2+O2

B. 2RbNO3→ 2Rb+2NO2+O2

C. 2RbNO3→2RbNO2+O2

D. Rb(NO3)2 → Rb(NO2)+O2

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48. The wavelength of radiation emits in Balmer series is:

A. Between 4000 A° and 8000A°

B. 10000< lamda <18000 A°

C. lamda<8000 A°

D. None of the above

49. Friedel crafts reactions classified as:

A. Halogenation and hydrogenation

B. Alkylation and Accylation

C. Halo-hydrogenation

D. Nitration

50. The formula of Sandhur is:

A. Pb3(OH2).(CO3)2

B. PbO4

C. PbO

D. Pb2O4

51. The atoms of the same elements, which have the same atomic number but different mass number are called:

A. Isobars

B. Actanids

C. Allotrops

D. Isotopes

52. In haber process:

N2+3H2 harr 2NH3

Increasing the pressure favours:

A. The forward reaction

B. Reverse reaction

C. Neither reaction

D. All of the above

53. Ksp is called:

A. Solubility Product

B. Concentration

C. Equilibrium

D. Constant concentration

54. Thermodynamics is the study of heat energy and _____ energy relationship.

A. Entergy of activation

B. Kinetic energy

C. Chemical

D. Ionization

55. The law of conservation of energy is also called:

A. Second law of thermodynamics

B. First law of thermodynamics

C. 3rd law of thermodynamics

D. None of the above

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56. Rb is an element in the same group of Li and Na; which one is true about Rb?

A. Reacts slowly with H2O

B. Forms an insoluble hydroxide

C. It is liberated at Cathode during the electrolysis of its salt in aqueous solution

D. Forms a sulphateRb2SO4

57. Which statement concerning energy change is false?

A. Electrical energy is produced by the reaction in a battery

B. Light energy is absorbed during photosynthesis

C. Heat energy is absorbed when Sodium reacts with water

58. Electrolysis of dil.H2SO4 gives:

CATHODE	ANODE
A) H_2	SO_2
B) H_2	O_2
C) O_2	H_2
D) SO_2	H_2

A. A

B. B

C. C

D. D

59. Carbohydrate contains:

A. C and H only

B. C, H and N only

C. C, H and O only

D. All of the above

60. An oxide of titanium contains 60% Ti. What is its empirical formula for this oxide?

A. TiO

B. Ti2O3

C. TiO2

D. Ti2O6

PHYSICS

61. If the distance between two charged bodies is doubled and the charge is also doubled the interaction force will be:

A. Half

B. Four times

C. Unchanged

D. Doubled

62. The capacitance of a capacitor depends upon:

A. Distance between the plates

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B. Area of the plates

C. Dielectric used between the plates

D. All of the above

63. If a radio and a bulb each of resistance 3Ω are connected in series to a 12 volts battery, the potential difference across each will be:

A. 3V

B. 6V

C. 12V

D. 9V

64. Choose the relation which is not correct:

A. Ampere=coulomb per second

B. Volt=coulomb per joule

C. Ohm=volt per ampere

D. 1 coulomb=charge of 6.24×10^{18} electrons

65. A particle of charge 'q' and mass 'm' moving with a velocity 'v' enters perpendicularly in a uniform magnetic field 'B' describes a circular path. What will be the radius of that path?

A. mv/Bq

B. mqv/B

C. mvB/q

D. mqB/v

66. Light of wavelength 15×10^{-7} m consists of photons whose energy is:

A. 1.7×10^{-15}J

B. 4×10^{-19} J

C. 1.1×10^{-37}J

D. 1.1×10^{-48}J

67. The electrons, structure of atom, are not involved in the emission of:

A. Spectral lines

B. Photoelectrons

C. Gamma rays

D. X-rays

68. When electron in hydrogen atom jumps from a higher orbit to the fifth orbit, the set of lines emitted are called:

A. Bracket Series

B. Paschen Series

C. Lyman Series

D. Pfund Series

69. Why gamma radiations are used to kill bacteria to sterilize surgical equipment etc?

A. Because they are chargeless

B. Because they have no mass

C. Because of its penetration

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D. Choice 3

70. The name of the most useful Tracer is:

A. Na-34

B. I-31

C. C-14

D. C-12

71. A wire of Uniform Cross sectional area has a length of 10m, a resistance of 2Ω and resistivity of $2x10^{-7}\Omega.cm$. The cross sectional area in m^2 is:

A. $2x10^{-4}$

B. $1x10^{-5}$

C. $0.5x10^{-5}$

D. $1x10^{-4}$

72. The power of a load is one watt if:

A. 1 ampere current flows through a conductor when a potential difference of 1 volt is applied

B. 0.1 ampere current flows when a potential difference of 0.1 volt is applied

C. 1 ampere current flows when a potential difference of 10 volts is applied

D. 10 ampere current flows when a potential difference of 10 volts is applied

73. If the rate of cost is Rs.2 per kWh then a 500 watt kettle operating for 30 minutes will cost:

A. Rs. 0.50

B. Rs. 0.075

C. Rs. 0.1

D. Rs. 1.50

74. An AC voltage measures Vo volts in a circuit. Its peak value is given by:

A. Vo

B. $Vo/\sqrt{2}$

C. $\sqrt{2}/Vo$

D. $\sqrt{2}$

75. A stationary photon has mass:

A. $1.67252x10^{-27}$ kg

B. $9.1x10^{-31}$ kg

C. Infinite

D. Zero

76. A battery of 6 volts is connected to two resistors of 30Ω and 20Ω joined together in series. Find the current through the circuit.

A. 12A

B. 1.2A

C. 36A

D. 150A

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77. An electric kettle of 1000 watts boils a certain quantity of water in 8 minutes. How much heat has been generated for boiling?

A. 4000J

B. 480000J

C. 4800J

D. 5000J

78. The current passes through all the resistors combined in a series should be _____.

A. Same

B. The sum of the current passes from each resistor

C. Potential difference over the value of equivalent resistance

D. Both A and C

79. The red line of the Balmer series originates in a transition between levels having quantum number:

A. 2 and 1

B. 3 and 1

C. 3 and 2

D. ∞ and 1

80. Solid state detector basically is:

A. A PNP transistor

B. A NPN transistor

C. A PN-Junction Diode

D. A light emitting diode

81. Michelson's interferometer is usually used to determine the:

A. Wavelength of light

B. Velocity of light

C. Frequency of light

D. Dispersion of light

82. How is a image, formed by a convex lens, affected if the upper half of the lens is covered with a paper?

A. The brightness of the image is reduced

B. The upper half of the image is cut off

C. The size of image becomes one half

D. No effect at all

83. A convex gives a virtual image only when the object is placed:

A. Between focus and centre of curvature

B. Between focus and optical center

C. Beyond 2f

D. Between f and 2f

84. If the pressure is increased, the melting point of ice:

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A. Decreases

B. Increases

C. Remains unchanged

D. None of the above

85. The process in which the pressure of system remains constant is called:

A. Isothermal process

B. Isochoric process

C. Isobaric process

D. Adiabatic process

86. Under Adiabatic compression the increase in temperature of a gas is proportional to:

A. The pressure to which the gas is compressed

B. Ratio of the two specific heats

C. Universal gas constant

D. Work done on the gas

87. A 50 mh coil carries a current of 2A. The energy stored in its magnetic field is:

A. 0.05J

B. 0.1J

C. 10J

D. 100J

88. In a amplitude modulation the waves that superimpose on electromagnetic waves are:

A. Water waves

B. Sound waves

C. Light waves

D. Material waves

89. Quality of X-rays depends upon:

A. Current of filament

B. Accelerating voltage

C. Material of target

D. Both accelerating voltage and material of the target

90. According to Bohr's theory of hydrogen atom, the angular momentum of electron in an allowed orbit is:

A. nh/π

B. $nh/2\pi$

C. $n/2\pi h$

D. $2\pi h/n$

ENGLISH

91. In order for people who spoke different languages to engage in trade

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_____, they often developed a simplified language called pidgin.

A. With each the other

B. With each to the other

C. With each another

D. With each other

92. The two main _____ are permanent magnets and electromagnets.

A. Kinds of magnets

B. Kind of magnets

C. Kind magnets

D. Kinds magnets

93. The supreme court does not hear a case u less _____, except those involving foreign ambassador.

A. A trial

B. Already tried

C. It already trying

D. It has already been tried

94. HOOF : HORSE (Analogies):

A. Lion : mane

B. Paw : cat

C. Pace : foot

D. Bicycle : Rider

E. Dog : Prairie

95. SHIP : DOC (Analogies)

A. Car : Park

B. Wheel : Steer

C. Horse : Ride

D. Raft : Float

E. Truck : Drive

96. EXPEDIENT (Synonym):

A. Scapegoat

B. Heterogeneous

C. Suitable

D. Apathetic

97. VORACIOUS (Synonym)

A. Reject

B. Excitable

C. Honest

D. Greedy

E. Inclusive

98. HAMPER (Antonym):

A. Ogle

B. Clarify

C. Assist

D. Uncover

E. Ventilate

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99. GAUNT (Antonym):

A. Rotund

B. Garish

C. Nervous

D. Stylish

E. Valiant

100. UNWARY (Antonym):

A. Alert

B. Sad

C. Angry

D. Eager

E. Dutiful

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ANSWER KEY

Q NO	ANS KEY	Q NO	ANS KEY	Q NO	ANS KEY	NO	ANS KEY
1	D	26	C	51	A	76	B
2	D	27	A	52	A	77	C
3	C	28	D	53	A	78	C
4	C	29	C	54	B	79	B
5	C	30	C	55	B	80	C
6	C	31	C	56	D	81	C
7	D	32	D	57	D	82	B
8	B	33	B	58	A	83	C
9	D	34	C	59	C	84	B
10	C	35	C	60	B	85	C
11	D	36	C	61	C	86	B
12	D	37	B	62	C	87	B
13	C	38	B	63	C	88	C
14	B	39	D	64	B	89	B
15	D	40	C	65	A	90	B
16	B	41	B	66	A	91	D
17	B	42	D	67	C	92	A
18	B	43	C	68	C	93	D
19	A	44	C	69	C	94	B
20	C	45	D	70	B	95	A
21	D	46	D	71	A	96	C
22	A	47	D	72	C	97	D
23	B	48	A	73	A	98	C
24	B	49	C	74	A	99	A
25	C	50	D	75	B	100	A

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